TECHGEN PROUDLY PRESENTS

Session 2026-27 · For registered teams

DITE Hackathon 2026-27

Participant Guidelines

Build a web or mobile app that uses AI to solve a real problem in healthcare, education, agriculture, cybersecurity or public service.

Team event · up to 3 members · 5 hours

DAY 1	**Friday 23 Oct 2026, Opening Ceremony 9:00 AM**
DURATION	**5 hours**
NAMAZ BREAK	**1:00 PM to 2:30 PM**
DAY 2	**Saturday 24 Oct 2026, 4:00 PM, then Sufi Night**
VENUE	**Auditorium, DITE, Bypass, near Kababjee's, Hyderabad**

About this guide

The clock is now running. Over the next 5 hours your team will build a working web or mobile application that uses AI to solve a real problem in one of the five themes revealed at the Opening Ceremony.

This guide covers everything you need from here to submission: the themes, what your project must include, how to push your code, deploy your app and record your demo, how to submit through the Google Form, and how the judges will score your work.

Read it together as a team before you start coding, and keep it close throughout the day. Most teams that lose marks do so because of missed submission rules, not weak ideas. Use the final checklist at the end before your team lead presses Submit.

1. Themes and the AI requirement

Every project must pick one of the five themes below, and AI must do real work in the solution. A chatbot bolted onto an unrelated app will not count.

Theme	Example problems to get you thinking
Healthcare	Symptom triage in Urdu or Sindhi, reading prescriptions, skin or X-ray image screening, appointment and medicine reminders for elderly patients
Education	Personalised quizzes, automatic grading of short answers, a tutor for local-language learners, early warning for students at risk of dropping out
Agriculture	Crop disease detection from leaf photos, irrigation advice from weather data, market price forecasting for farmers
Cybersecurity	Phishing email or SMS detection, fake news or scam link checking, password strength and breach awareness tools
Public service	Complaint routing for municipal issues, document and form assistants, traffic or waste reporting with image recognition

These are only examples. You are free to choose your own problem as long as it clearly fits one theme.

What counts as “using AI”

- Training or fine-tuning your own model (classification, detection, segmentation, forecasting, NLP)
- Using a pre-trained model, for example from Hugging Face, TensorFlow Hub or PyTorch Hub
- Calling an AI API such as a large language model, speech-to-text, vision or translation service
- Combining several of the above in one pipeline

> Whatever you use, the judges should be able to see where the AI sits in your app and why the problem needs it. Be ready to explain this in your video and in your README.

2. What you need to build

You must build either a web application or a mobile application. You may use any language, framework or tool you are comfortable with.

Common choices (none of these are required)

Area	Options
Web	React, Next.js, Vue, Angular, Django, Flask, FastAPI, Node.js with Express, Laravel, Streamlit or Gradio
Mobile	Flutter, React Native, Kotlin or Java for Android
AI	Python with PyTorch, TensorFlow, scikit-learn or OpenCV; Hugging Face models; LLM and vision APIs
Database	Firebase, Supabase, MongoDB, PostgreSQL, MySQL or SQLite

Minimum requirements for every project

1. The app must run. A live link or installable APK that opens and works is required.
2. The AI feature must be working inside the app, not shown separately in a notebook.
3. The main user flow should work from start to finish without crashing.
4. The interface should be usable by the people you are building for. Think about language, simple screens and slow internet.

A few practical points

- Never put API keys, passwords or tokens directly in your code. Use environment variables and add your `.env` file to `.gitignore`.
- If you use a paid API, check your free quota early. A demo that stops working because credits ran out will be judged as not working.
- Large model files usually cannot be pushed to GitHub. Host them on Hugging Face, Google Drive or your deployment platform and load them from there.
- If you use a public dataset, mention its name and link in your README.

3. GitHub repository

All code must be pushed to a public GitHub repository, and that link goes in the Google Form.

- Make the repository **public**. Judges will not request access to private repos.
- Push regularly during the hackathon rather than once at the end. Your commit history is a simple way for us to see that the work was done during the event.
- Commits from all team members are encouraged.
- Do not push any code changes after the submission deadline. Commits made after the deadline may lead to disqualification.
- Remove secrets before pushing. If you accidentally push a key, revoke it immediately and create a new one.

Your README must include

1. Project name and team name with member names

2 Theme chosen and the problem you are solving, in two or three sentences
3 How AI is used: model or API name, what it takes as input and what it returns
4 Tech stack
5 Steps to run the project locally
6 Live deployment link or APK link
7 Demo video link
8 Datasets, pre-trained models and other resources used, with credit to their authors

A screenshot or two in the README helps judges understand your project quickly.

4. Deployment

Web apps must be live on a public URL, and mobile apps must be shared as an installable APK.

For web applications

Deploy on any online platform you like. Free options that teams commonly use include Vercel, Netlify, Render, Railway, Hugging Face Spaces, Streamlit Community Cloud, Firebase Hosting and GitHub Pages (for static frontends only).

- The link must open without a login wall for judges. If your app needs an account, create a demo account and put the username and password in the Google Form notes, not in your public README.
- Free servers often sleep after a period of inactivity. Open your link shortly before judging so it wakes up.
- Keep the app live until results are announced on Saturday 24 October.

For mobile applications

- Build a release or debug APK that installs on a normal Android phone.
- Upload the APK to Google Drive (or GitHub Releases) and set sharing to **Anyone with the link can view**.
- Test the APK on at least one real phone, not only on an emulator.
- If the app depends on a backend or API, make sure that backend is also deployed and running.

> Before submitting, open your link or install your APK from a different device and a different account. If it does not work there, it will not work for the judges.

5. Demo video

Record a demo video of 3 to 5 minutes, upload it to Google Drive and share the link in the form.

Suggested structure

Part	What to show	Time
1	Introduce the team and the problem	~30 sec
2	Who the users are and why the problem matters	~30 sec
3	The app working end to end, as a real user would use it	2 to 3 min
4	The AI feature, explained in simple words	~1 min
5	Limitations and what you would build next	~30 sec

Technical rules

- Use a screen recorder with your voice over it. Free tools like OBS Studio or the phone's built-in recorder work well.
- Speak in English or Urdu, clearly and not too fast.
- Minimum resolution of 720p.
- Name the file like this: `TeamName_ProjectName_Demo.mp4`
- Set Drive sharing to **Anyone with the link can view**. Open the link in an incognito window to confirm it plays.

Show the real app. Slides alone, or mock screens that do not actually work, will be marked down.

6. How to submit

Only the team lead submits, once, through the Google Form shared by the organisers, before the 5 hours are up.

1. Push your final code to the public GitHub repository and check that the README is complete.
2. Deploy your web app and copy the live URL, or upload your APK to Drive and copy its share link.
3. Upload your demo video to Google Drive and copy its share link.
4. Open every link in an incognito window to make sure it works without your login.
5. Fill the Google Form with the details below and press Submit.
6. Keep the confirmation email or take a screenshot of the confirmation page.

The form will ask for

Field	What to enter
Team name	As registered
Team lead name, email and phone	Contact person for the team
Team members	Full names of all members
Project title	Short and clear
Theme	Healthcare, Education, Agriculture, Cybersecurity or Public service
Problem statement	Two or three sentences
AI used	Model or API name and what it does in your app
GitHub repository link	Public repo URL
Deployment link or APK link	Live web URL, or Drive / GitHub Releases link to the APK
Demo video link	Google Drive link, set to anyone with the link
Demo login (if needed)	Test username and password for judges

If you notice a mistake after submitting, contact the organisers before the deadline. Do not submit the form a second time unless they ask you to.

7. How projects will be judged

Projects are scored out of 100, and the largest share goes to whether the AI genuinely solves the problem.

Criterion	Marks	What judges look for
Use of AI	25	AI is central to the solution, used sensibly, and works on real inputs
Problem relevance and impact	20	A real problem in the chosen theme, with clear users who would benefit
Working product	20	Deployed link or APK works, main flow runs without crashing
Innovation	15	A fresh idea or a smart new angle on a known problem
Design and usability	10	Clean, simple interface suited to the target users
Code quality and documentation	5	Readable code, sensible structure, complete README
Demo video and presentation	5	Clear explanation, real working demo, within time
Total	**100**	

Every team pitches its project to the judges after the 5 hours end, followed by questions. Results are announced at the Closing Ceremony on Saturday 24 October at 4:00 PM. The judges' decision is final.

Organisers may adjust these weights before the event. Any change will be announced to all teams at the same time.

8. Code of conduct and originality

We expect honest work and respectful behaviour from every participant.

Originality

- The project must be built today, during the hackathon. You may have planned your idea beforehand, but the code should be written during the event.
- Open-source libraries, templates, pre-trained models and public datasets are allowed, as long as you credit them in your README.
- AI coding assistants (ChatGPT, Claude, Copilot and similar) are allowed. You are still responsible for understanding every part of your code and should be able to explain it when asked.
- Submitting someone else's project, or an old project of your own with minor changes, is not allowed.

Behaviour

- Treat other teams, volunteers, mentors and judges with respect.
- Harassment or discrimination of any kind will not be tolerated.
- Look after the campus facilities and leave your workspace clean.
- Follow instructions from the organising team at all times.

Data and ethics

- Do not use real patient, student or citizen data unless it is from a public, properly anonymised dataset.
- Cybersecurity projects must be defensive. Tools that attack, hack or harm real systems or people are not permitted.
- Be honest in your demo about what your model can and cannot do.

9. Reasons for disqualification

A team may be disqualified for any of the following:

- Submitting after the deadline, or not submitting through the Google Form
- Missing any of the three required links: GitHub, deployment or APK, and demo video
- A private GitHub repo, or Drive links that judges cannot open
- No AI component, or AI that plays no real role in the solution
- A project outside the five themes
- Plagiarised code or a project made before the hackathon
- Pushing code changes after the deadline
- Misconduct during the event

10. Final checklist

Tick these off before the team lead presses Submit.

- ☐ Project fits one of the five themes
- ☐ AI is working inside the app
- ☐ Code pushed to a public GitHub repo with a complete README
- ☐ No API keys or passwords in the repo
- ☐ Web app is live, or APK installs on a real phone
- ☐ Demo video is 3 to 5 minutes and shows the real app
- ☐ All Drive links set to anyone with the link can view
- ☐ Every link tested in an incognito window
- ☐ Google Form submitted before the 5 hours are up
- ☐ Confirmation saved

Tips from past hackathons

- Pick a small problem and solve it properly. A narrow app that works beats an ambitious one that half works.
- Get a basic version deployed early, then improve it. Deployment problems at the last hour are the most common reason teams miss the deadline.
- Use a pre-trained model first. Train your own only if you have the time and the data.
- Split the work clearly across your three members: one on the AI part, one on the app, one on deployment and the video.
- Talk to the mentors. They are there to help you get unstuck.
- Five hours goes quickly. Record your demo video well before the end, not in the last ten minutes.

Contact

For any question about the rules or submission during the day:

- Hackathon coordinators: **Sir Yasir Nawaz** and **Sir Waqas Aziz**
- TECHGEN volunteers in the auditorium

> **Good luck.** Build something you would be proud to show your family and the people it is meant to help.